

Quarterly GDP Forecasting Report: Sector-Wise Analysis and Model Comparison

Objective

This report aims to provide a comprehensive forecast of India's Gross Domestic Product (GDP) for the next four quarters using historical quarterly data. The focus is on understanding economic trends through a sector-wise breakdown—covering the Primary, Secondary, and Tertiary sectors—and identifying the most effective forecasting method based on past performance. The analysis also explores how these sectors individually contribute to the overall economic growth, enabling more informed interpretations and decisions.

Data Overview

The dataset consists of quarterly GDP figures spanning multiple years, including detailed values for three major economic sectors: Primary, Secondary, and Tertiary. The Primary Sector includes agriculture, forestry, fishing, and mining. The Secondary Sector comprises manufacturing, electricity generation, and construction. The Tertiary Sector encompasses services, trade, transport, and financial activities. Each record in the dataset represents one fiscal quarter, starting from 2011. This structure allows for capturing both long-term trends and seasonal effects in the Indian economy.

Forecasting Methods

To generate reliable GDP forecasts, five different models were applied to the data. The first was Linear Regression, a basic method that assumes GDP changes steadily over time in a linear fashion. While simple, this model often fails to capture real-world complexities like economic shocks or seasonal variations.

The second model, ARIMA (Auto Regressive Integrated Moving Average), uses historical data and prior forecast errors to project future values. It is more adaptable than linear regression but does not inherently capture seasonal patterns.

The Holt-Winters model, also known as triple exponential smoothing, is particularly suitable for datasets with both trends and seasonality. It adjusts its predictions based on past trends and repeated seasonal fluctuations, making it useful for quarterly economic data.

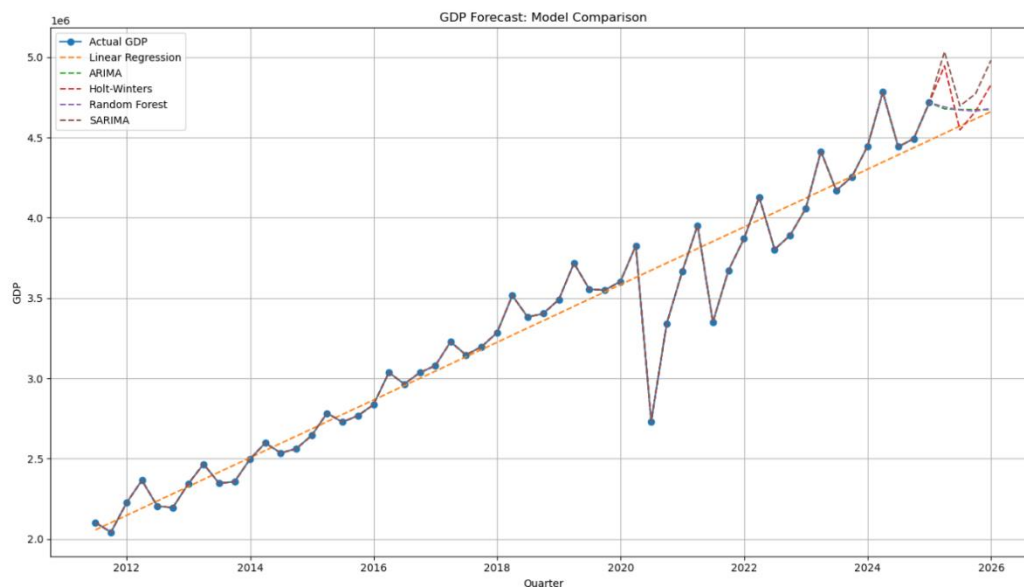
The Random Forest model is a machine learning approach that creates many decision trees from past data. It attempts to capture complex, non-linear relationships, although it requires enough historical data to train effectively.

The SARIMA model (Seasonal ARIMA) extends ARIMA by incorporating seasonality directly. It is specifically designed for time series data like GDP, which exhibits both short-term trends and seasonal patterns. Among the models used, SARIMA aligns best with the characteristics of quarterly economic data.

Model Evaluation Metrics

To evaluate each model's accuracy, several statistical metrics were used. RMSE (Root Mean Squared Error) measures how much the predicted values deviate from actual values, giving more weight to larger errors. MAE (Mean Absolute Error) calculates the average size of the errors. MAPE (Mean Absolute Percentage Error) expresses the error as a percentage, offering easy interpretation across values of different magnitudes. Median AE (Median Absolute Error) shows the middle value of the errors and is less affected by outliers. R^2 Score indicates how much of the variation in GDP can be explained by the model, where a score closer to 1 is ideal.

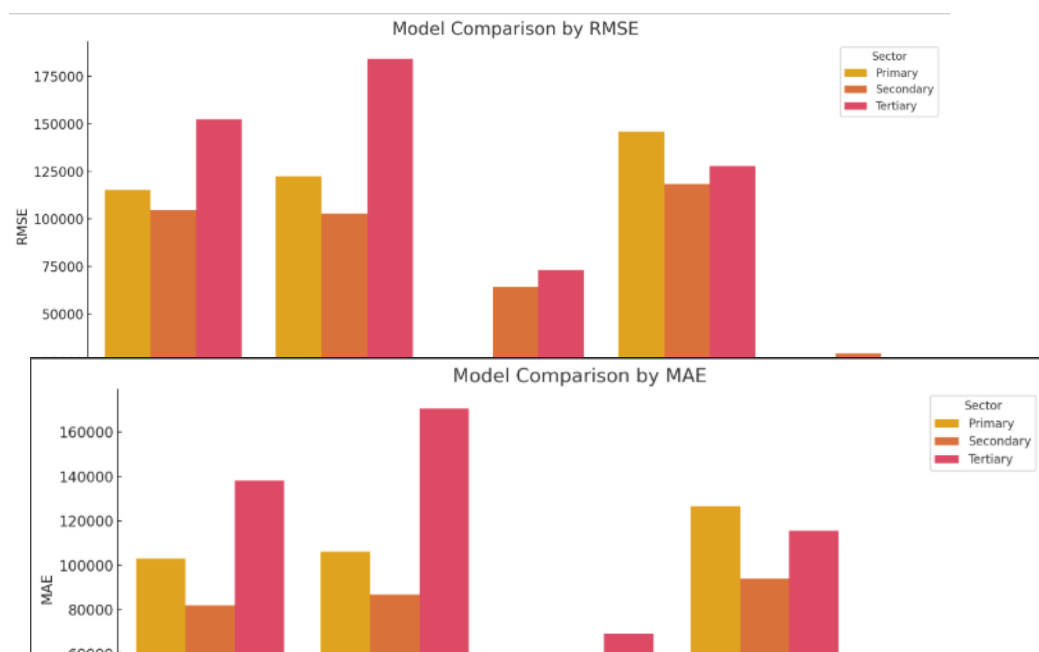
The results were visualized using bar charts for RMSE, MAE, and MAPE, comparing performance across sectors (Primary, Secondary, and Tertiary) and models. These graphs clearly highlighted the superior performance of SARIMA in all sectors.

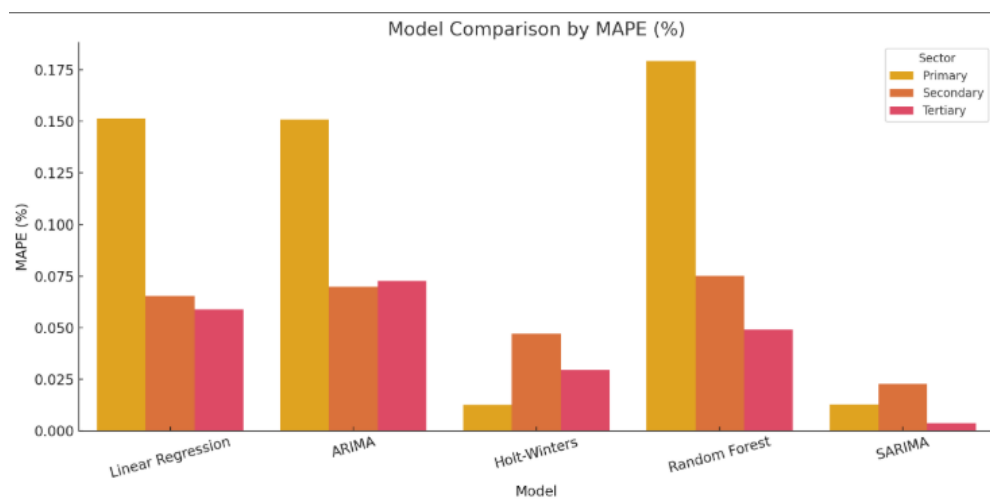


Model Performance Across Sectors

The performance of each forecasting model was assessed individually for the Primary, Secondary, and Tertiary sectors. Across all three sectors, the SARIMA model consistently delivered the most accurate forecasts. It demonstrated the lowest errors in terms of RMSE, MAE, and MAPE, and its predictions were closest to the actual GDP values observed during the test period. Linear Regression, by contrast, showed the poorest performance, as it could not adapt to the seasonality and variability inherent in economic data.

Graphs comparing RMSE, MAE, and MAPE across models and sectors are included to visually demonstrate this conclusion. Each chart highlights how the SARIMA bars are consistently the lowest, indicating better performance.





	Model	RMSE	MAE	MAPE (%)	Median AE \
0	Linear Regression	305573.160211	262158.825264	559.322223	215482.776486
1	ARIMA	247287.372884	201654.060829	428.297567	198996.762178
2	Holt-Winters	158487.759170	153604.389975	333.447084	162962.430117
3	Random Forest	272157.535815	233015.025000	496.589709	233959.900000
4	SARIMA	70654.310055	62691.325418	137.552858	60945.204433

	R2 Score
0	-3.492996
1	-1.942453
2	-0.208640
3	-2.564071
4	0.759795

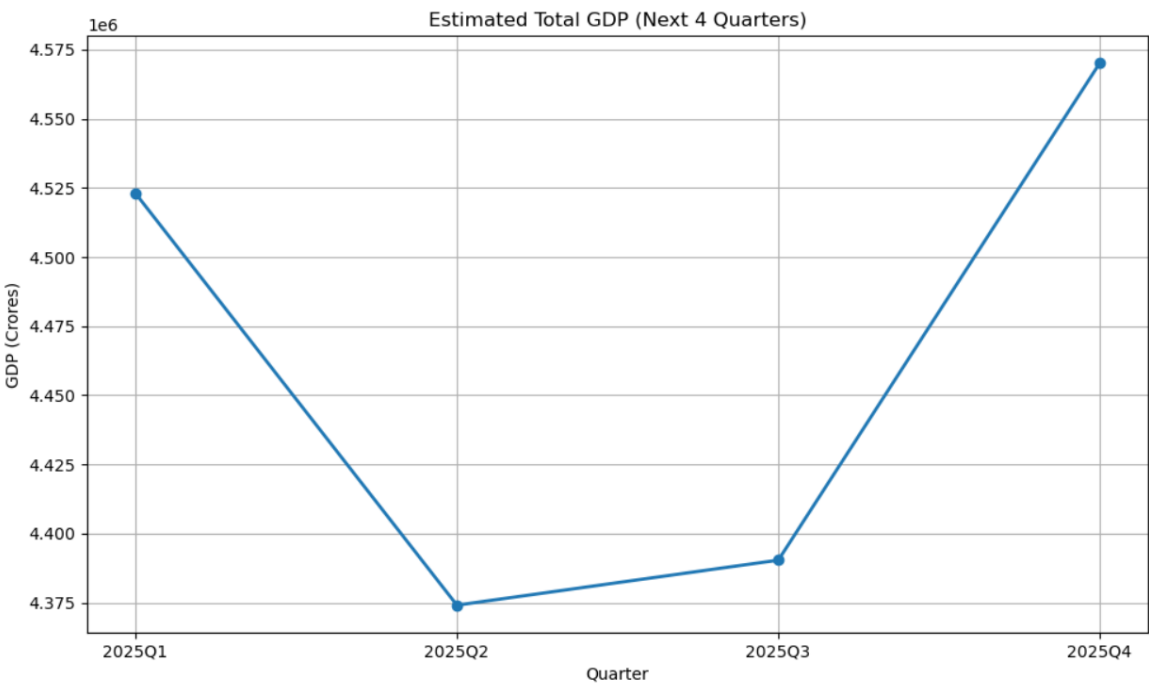
Sector-Wise GDP Forecasts

Using the SARIMA model, forecasts for each of the three sectors were generated for the next four quarters. The forecasted values were then summed to produce an overall GDP estimate. The projections are as follows:

Quarter Primary Sector Secondary Sector Tertiary Sector Estimated Total GDP

2025 Q1	₹7,89,873	₹13,44,641	₹23,88,533	₹45,23,047
2025 Q2	₹6,83,202	₹12,22,466	₹24,68,412	₹43,74,080
2025 Q3	₹5,87,529	₹12,32,663	₹25,70,182	₹43,90,374
2025 Q4	₹8,95,173	₹12,39,845	₹24,35,134	₹45,70,152

These forecasts were also visualized using a line chart to depict the expected trend in GDP over the upcoming quarters. The chart showed minor fluctuations, with peak growth expected in Q1 and Q4, aligning with seasonal economic activity patterns.



Summary and Conclusion

The forecasting exercise demonstrates that SARIMA is the most appropriate and accurate model for projecting India’s quarterly GDP across sectors. It consistently outperforms other models by effectively capturing the seasonal patterns and underlying trends in economic data. Sector-wise forecasting further adds value by revealing the unique contributions of each economic segment to the total GDP.

Forecasts for the next year indicate that the economy will grow at a moderate pace, with seasonal peaks driven by agriculture and services. These projections are valuable for policymakers in budget planning, economists in strategic analysis, and businesses in preparing for market fluctuations.

The findings validate the importance of using sector-level data and advanced time series models when analysing economic indicators. This approach not only improves forecasting accuracy but also provides clearer insights into the structural dynamics of the economy.